BTS–IV(S)–04.21–0252 Reg. No. [| | | | | | | |]

A

B.Tech. Degree IV Semester Supplementary Examination April 2021

ME 15-1402 METROLOGY AND INSTRUMENTATION

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) What are Systematic errors? Can we eliminate these errors?
   (b) Discuss the effect of supports on deformation of measuring bars.
   (c) What do you mean by 'hole basis system'? Briefly explain.
   (d) Explain the importance of an optical flat.
   (e) State the essential properties of light waves for interference.
   (f) What do you mean by lay? Give symbols of different types of lay on surface.
   (g) What is static calibration? Explain.
   (h) Distinguish between active and passive transducer.
   (i) What is a sound level meter? What are its basic parts?
   (j) What are the various pollutants in air? Mention different methods used to find out air quality.

PART B

(4 × 10 = 40)

II. (a) For a measuring instrument, discuss: (5)
   (i) Sensitivity (ii) Magnification (iii) Calibration (iv) Precision (v) Traceability
   (b) Which agency maintains standards internationally? Explain why material standards are replaced by non-material standards. Use examples to substantiate your points. (5)

OR

III. (a) Compare Clearance fit and Interference fit. Give examples. (5)
   (b) Explain the significance of Tolerance and Interchangeability in mass production. (5)

IV. (a) Write a short note on sprit level. (5)
   (b) In the measurement of surface roughness, heights of 20 successive peaks and valleys measured from datum are as follows: (5)
   45, 25, 40, 25, 35, 16, 40, 22, 25, 34, 25, 40, 20, 36, 28, 18, 20, 25, 30, 38.
   If these measurements were made over a length of 20 mm, determine the C.L.A and R.M.S values of the surface.

OR

V. Explain the working of following instruments with neat sketch (10)
   (i) NPL flatness type interferometer (ii) Laser beam interferometer

VI. What are the different methods of correction for interfering and modifying inputs? Explain with examples. (10)

OR

VII. (a) Explain the generalized measurement system with block diagrams. Give an example. (6)
   (b) Describe the working of potentiometer transducer as zero order instrument. (4)

VIII. (a) Explain the working of Scintillation Counter with neat sketches. (5)
   (b) Discuss five laws of thermocouple. (5)

OR

IX. (a) Explain any temperature measurement process without involving physical contact, in detail. (5)
   (b) What do you mean by temperature compensation in strain gauge? Why is it required? (5)